MIL-C-36023A
23 May 1975
SUPERSEDING
MIL-C-36023
5 June 1959

MILITARY SPECIFICATION

CALAMINE LOTION, PHENOLATED AND MENTHOLATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Calamine Lotion, Phenolated and Mentholated.

1.2 Classification. The lotion covered by this specification shall be the following sizes (See 6.2):

Size 1	2 fl oz.
Size 2	4 fl oz.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company Easton, Pa. 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Material shall be a uniform, homogeneous, grit-free, modified calamine lotion in accordance with the following formula:

Calamine	80 Grams
Zinc Oxide	80 Grams
Menthol	10 Grams
Liquefied Phenol	10 ml
Glycerin	20 ml
Bentonite Magma	400 ml
Calcium Hydroxide Solution, a sufficient quantiy, to make	1000 ml

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The ingredients shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2 Assay tolerances. The modified calamine lotion shall comply with the following assay tolerances.

3.2.1 Assay for phenol content. Phenol content shall be not less than 0.80 gram and not more than 1.00 gram in each 100 milliliters (ml) of lotion when tested as specified in 4.5.1.

3.2.2 Menthol content. The menthol content shall be 1.0% as determined from the formulation sheet.

3.2.3 Assay for total zinc oxide and bentonite plus ferric oxide content. The total zinc oxide content shall be not less than 14.50 grams and not more than 17.50 grams in each 100 ml of lotion, and the bentonite plus ferric oxide content after ignition shall be not less than 1.60 grams and not more than 2.00 grams in each 100 ml of lotion when tested as specified in 4.5.2.

3.2.4 Residue on ignition. The residue on ignition shall be not less than 16.10 grams and not more than 19.50 grams in each 100 ml of lotion when tested as specified in 4.5.3.

3.2.5 Stability of suspension. The lotion shall show neither more than 2 ml separation of supernatant liquid nor the presence of oil droplets when tested as specified in 4.5.4.

3.3. Workmanship. The material and its containers shall be free from any defects which detract from their appearance or impair their serviceability.

3.4 Delivery schedule. Not more than 6 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such recods relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of the modified calamine lotion produced from one formulation, subjected to the same compounding and manufacturing operations, and placed into the final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I Sampling

For Visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Bottle
Minor	II	2.5	Bottle

4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection.

Table II. Testing

Characteristic	Requirement paragraph	Test paragraph	Inspection level*	Minimum Sample
Phenol content	3.2.1	4.5.1	S-2	5
Menthol content	3.2.2	—	—	—
Total zinc oxide and bentonite plus ferric oxide content	3.2.3	4.5.2	S-2	5
Residue on ignition	3 2.4	4.5.3	S-2	5
Stability of suspension	3.2.5	4.5.4	S-2	5

*Where possible, the same sample shall be used for two or more tests.

4.4 Examination. The lotion shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Table III Classification of defects

Categories	Defects*
Major	
101	Lotion not free of grit.
102	Lotion not uniform or homogeneous.
103	Contents not free from foreign matter.
Minor	
201	Exterior of immediate container not free from contents or foreign matter

*Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

Note: Exercise adequate care in all volume measurements of the lotion, since the air entrapped when the lotion is agitated may result in significant errors in the measurements.

4.5.1 Phenol assay. Weigh accurately about 5.0 grams of the thoroughly shaken, modified calamine lotion and transfer to a 500-ml iodine flask. Add 125 ml of purified water and 50 ml of 0.1N bromine. Then add 5 ml of hydrochloric acid and immediately stopper the flask. Shake the mixture for 30 minutes and then allow to stand for 15 minutes in a dark place. Remove the stopper sufficiently to quickly introduce 5 ml of a 20 percent solution of potassium iodine, being careful that no bromine vapor escapes, and immediately restopper the flask. Shake the mixture thoroughly and rinse the stopper and the neck of the flask with a small portion of purified water, allowing the washings to flow into the flask. Add 1 ml of chloroform and shake the mixture. Titrate the liberated iodine with 0.1N sodium thiosulfate, using starch T.S. as the indicator, to a salmon-pink end point. Perform a blank titration with the same quantities of the same reagents and in the same manner. Each ml of 0.1N bromine is equivalent to 1.569 milligrams (mg) of phenol.

Calculations:

(a) $$\frac{\text{Weight of sample (Gram)}}{\text{Specific gravity of lotion}} = \text{Volume of sample (ml)}$$

(b) $$\frac{(\text{ml used in blank titration} - \text{ml used in sample titration}) \times 0.001569 \text{ Gram} \times 100}{\text{Volume of sample (ml)}} = \text{Gram of phenol/100 ml.}$$

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4.5.2 Assay for total zinc oxide and bentonite plus ferric oxide. Transfer a thoroughly mixed sample of lotion to a 10-ml volumetric flask, by means of a pipet, filling the flask to the mark. Then transfer the sample on to a filter paper in a Buchner funnel. Wash the flask with purified water and transfer the washings to the funnel. Wash the sample in the funnel with cold purified water until the filtrate is neutral to litmus paper. Transfer the filter paper with the residue to a 250-ml beaker, and add 50 ml of 1N sulfuric acid. Heat gently for a few minutes, remove from heat and stir until no further solution takes place. Filter into a 250-ml volumetric flask being sure to transfer all of the residue to the filter. Wash the residue with hot purified water until the filtrate is neutral to litmus paper. Cool and make up the combined filtrate and washings to the mark with the purified water. (Do not discard the filter paper and residue. Follow the procedure specified below* to determine the quantity of residue.) To a 50 ml aliquot, add 1.5 grams of ammonium chloride and one drop of methyl orange T.S. Titrate with 1N sodium hydroxide until the color of the indicator changes from pink to pale orange. It is preferable that stirring be accomplished with the aid of a mechanical or magnetic stirrer during titration. Each ml of 1N sulfuric acid is equivalent to 40.69 mg of zinc oxide.

*Place the filter paper and residue into a tared crucible and ignite to constant weight, using a Mekker burner.

(a) Total Zinc Oxide

$$\frac{(\text{ml 1N } H_2SO_4 - \text{1N NaOH}) \times 0.04069 \text{ Gram} \times 100}{\text{Volume of sample (ml)}} = \text{grams of zinc}$$

(b) Residue of Bentonite plus Ferric Oxide

$$\frac{\text{Weight of residue (Gram)} \times 100}{\text{Volume of sample (ml)}} = \text{grams of bentonite plus ferric oxide/100 ml.}$$

4.5.3 Residue on ignition. Weigh accurately about 10 grams of thoroughly mixed lotion into a tared crucible. Carefully heat and then ignite to constant weight using a Meker burner.

Calculation:

(a) $\frac{\text{Weight of sample (Grams)}}{\text{Specific gravity of lotion}} = \text{Volume of sample (ml)}$

(b) $\frac{\text{Weight of residue (Gram)} \times 100}{\text{Volume of sample (ml)}} = \text{Grams of total residue on ignition/100 ml.}$

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4.5.4 Stability of suspension. Shake the modified calamine lotion and transfer it to a 100-ml graduated cylinder having a ground glass stopper. Cylinder shall be 27 millimeters (mm) ± 3 mm in inside diameter and shall be graduated at 1.0 ml increments. Fill to the 90-ml mark with the lotion. Stopper the cylinder. Moderately shake the contained lotion for one minute and then allow the cylinder to stand undisturbed for 15 minutes.

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable paragraphs of PPP-C-186 and as specified herein.

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5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 2 fl. oz. (size 1) or 4 fl. oz. (size 2), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considereded acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each bottle shall be packaged as specified in paragraph 5.2.5 of PPP-C-00186.

5.2.2.3 Intermediate package. Twelve units of size 1 material shall be packaged as specified in 5.2.7.1 of PPP-C-186, and twelve units of size 2 material shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twelve intermediate packages of size one material or eight intermediate packages of size 2 material shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve intermediate packages of size 1 material shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186 or eight intermediate packages of size 2 material shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

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5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".
(c) For use as a protective agent to the skin and for relief of itching.
(d) Shake Well. Apply with fingers or cotton pad to itching skin. Dab, do not rub.
(e) Suitable warnings as recommended by F.D.A. for labeling of external preparations containing more than 0.5% carbolic acid. As a minimum, cautions against application to large body areas and against bandaging when applied to fingers and toes.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number

NOTE: (1) The national stock number shall not be proceeded by any prefix.
(2) Strip labeling may be used.
(3) Labeling adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with paragraph 5.5.3 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE"

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5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-111-1200	CALAMINE LOTION, PHENOLATED AND MENTHOLATED, 2 fl oz (59 ml)
6505-00-656-1466	CALAMINE LOTION, PHENOLATED AND MENTHOLATED, 4 fl oz (118ml)

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1389
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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